

Computational Wind Engineering

The Atmospheric Boundary Layer

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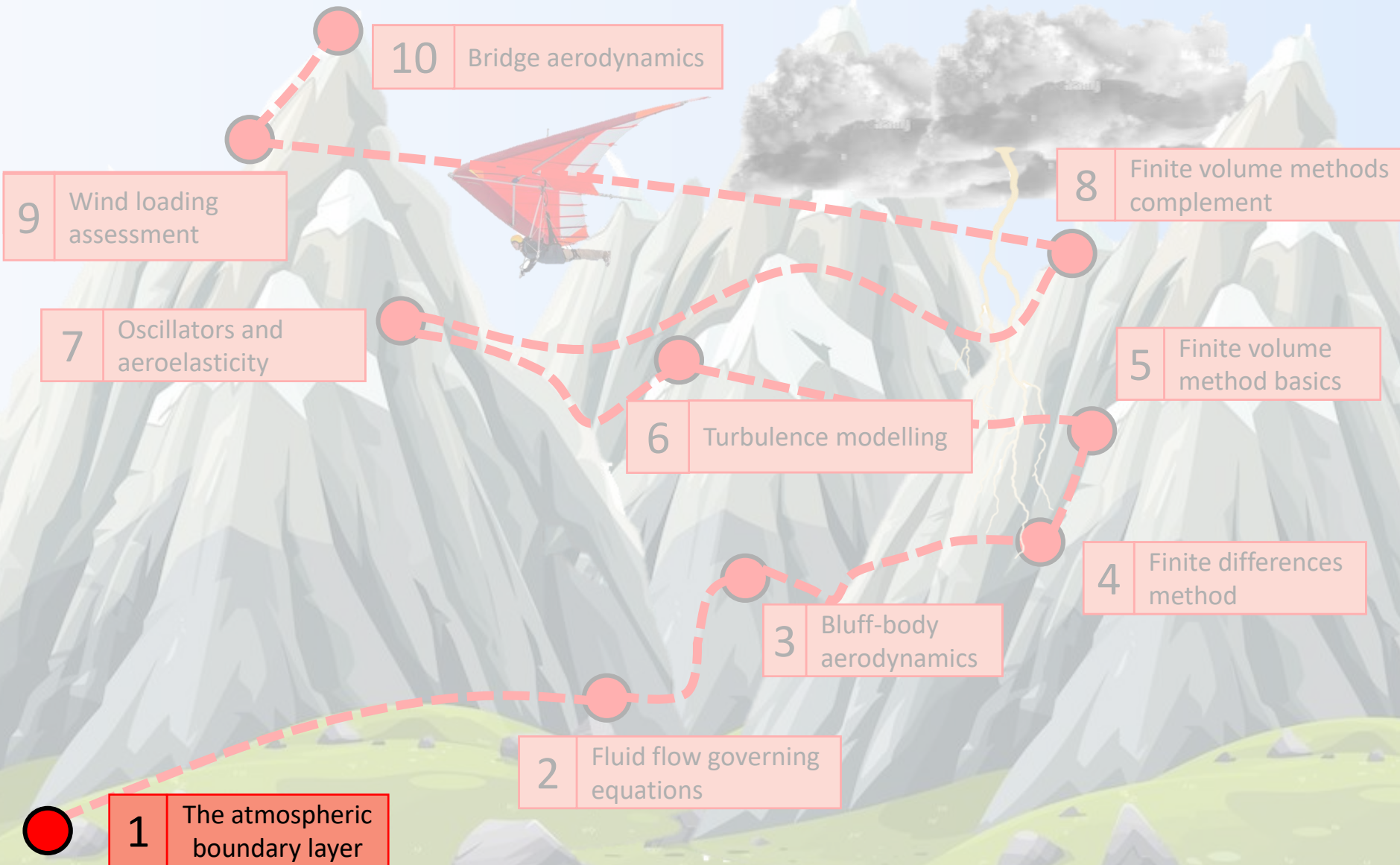
- Global circulation
- Davenport's chain of wind loading
- Wind profiles
- Random variables
- Turbulence metrics

Map of the course

Wind Engineering

Fluid Dynamics

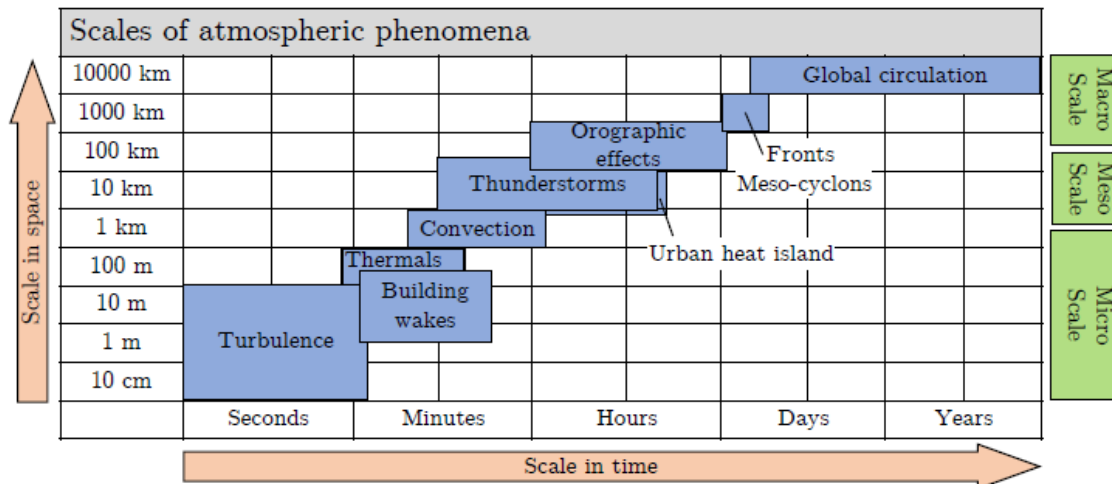
Numerical methods



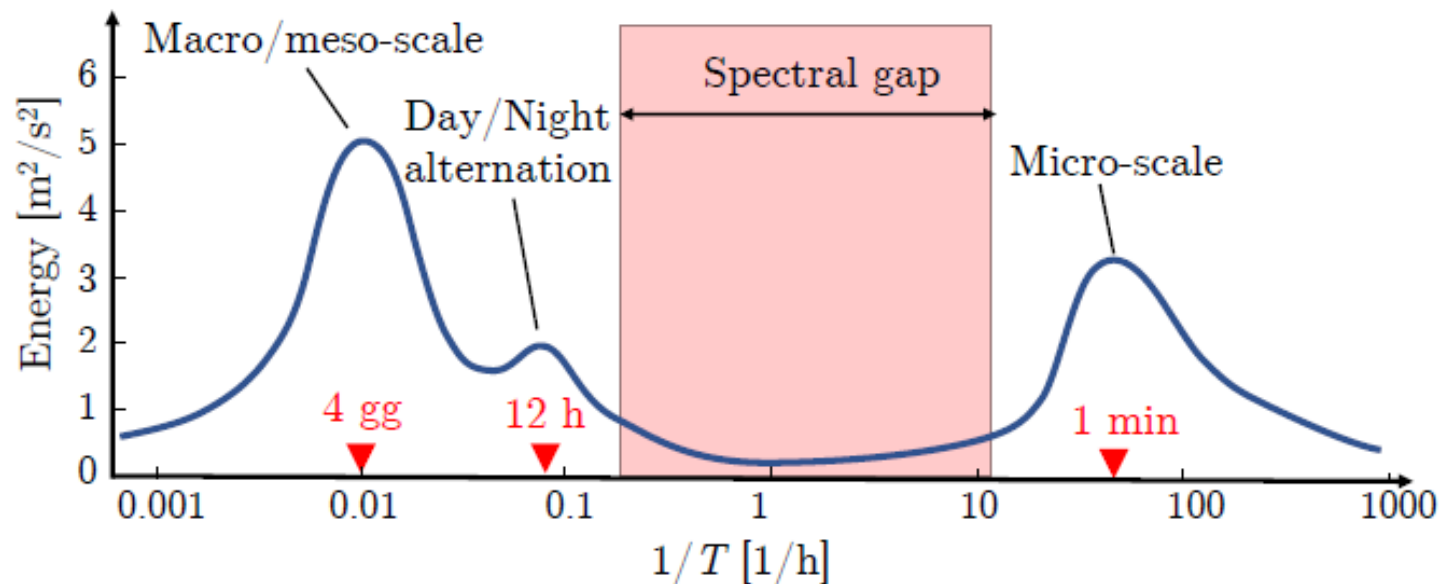
- will it be windy tomorrow between 2:00 and 3:00 PM?
- what will be the wind speed at a specific point in 2 min time?

Scales of atmospheric phenomena

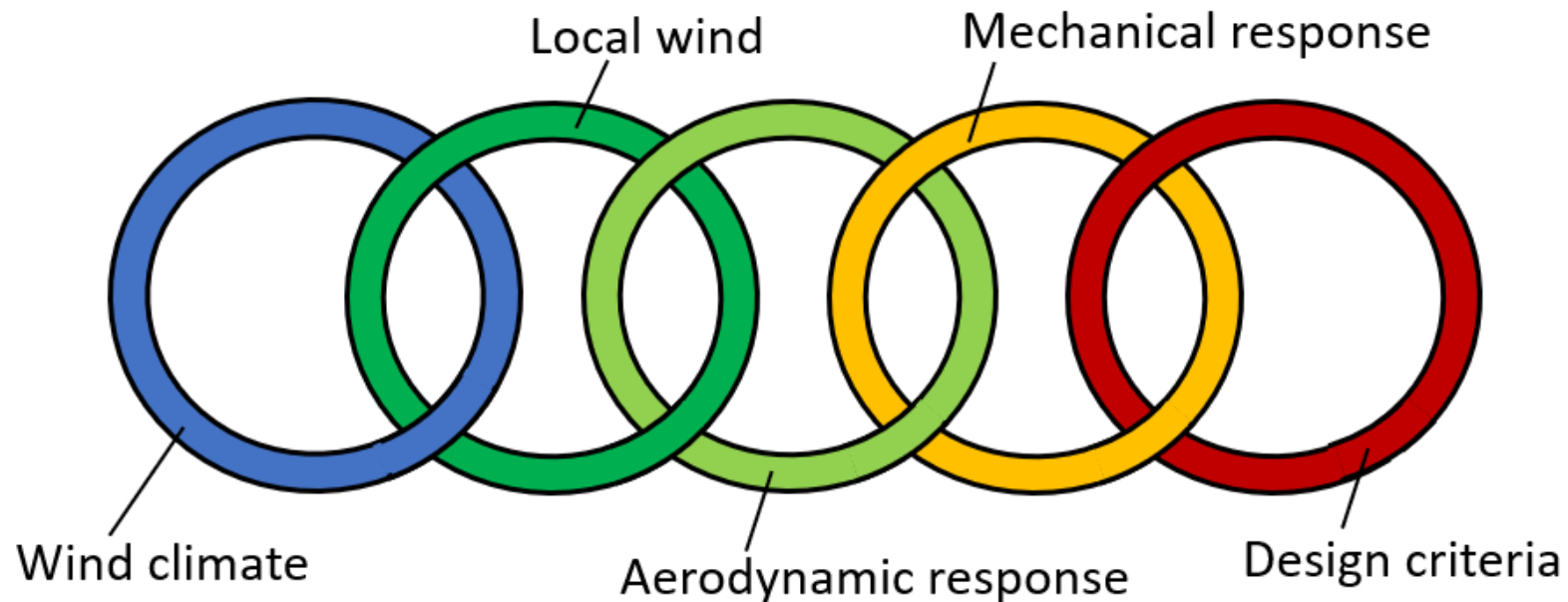
Scale in space (km)	Scale in time	Phenomena	Scale Category
10000	Months to Years	Global circulation	Macro Scale
1000	Days to Weeks	Orographic effects	Meso Scale
100	Hours to Days	Thunderstorms, Fronts, Meso-cyclons	Meso Scale
10	Minutes to Hours	Convection	Meso Scale
1	Seconds to Minutes	Thermals	Micro Scale
100	Minutes to Hours	Turbulence	Micro Scale
10	Seconds to Minutes	Building wakes	Micro Scale
1	Seconds to Minutes	Urban heat island	Micro Scale
100	Seconds to Minutes	-	Micro Scale
10	Seconds to Minutes	-	Micro Scale
1	Seconds to Minutes	-	Micro Scale
10	Seconds to Minutes	-	Micro Scale



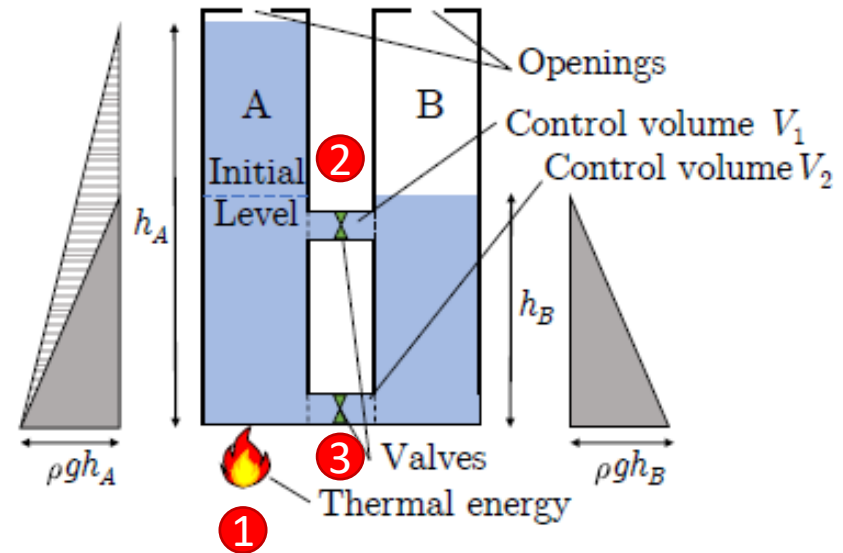
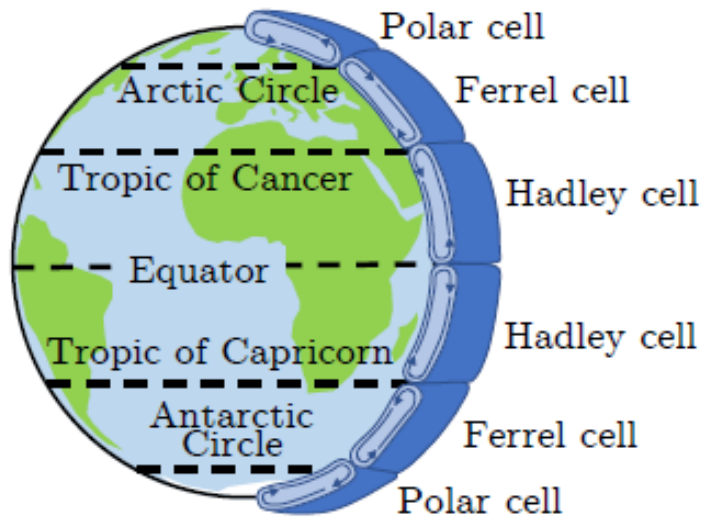
In 1957 Van der Hoven provided a plastic representation of the scales which affect wind speeds. In particular, he measured the power contained in wind at different time scales, leading to the so-called wind spectrum.



Thanks to the existence of the **spectral gap**, it is possible to study the effects of the wind climate and the local wind, separately. Davenport formalised the steps necessary for a general wind load study in his famous chain diagram.



At the planetary scale, **air circulation self-organises in convective cells**. The energy required for the air motion within these cells is provided by temperature gradients, which lead to pressure gradients.



A simplified scheme of the working principle of a convective cell is given by the scheme above. The mid-latitude cell is different from others, as it is a secondary one, actioned by the other two.

Numerous other aspects contribute to the wind climate, which we do not consider here. Two of particular interest are



Downburst

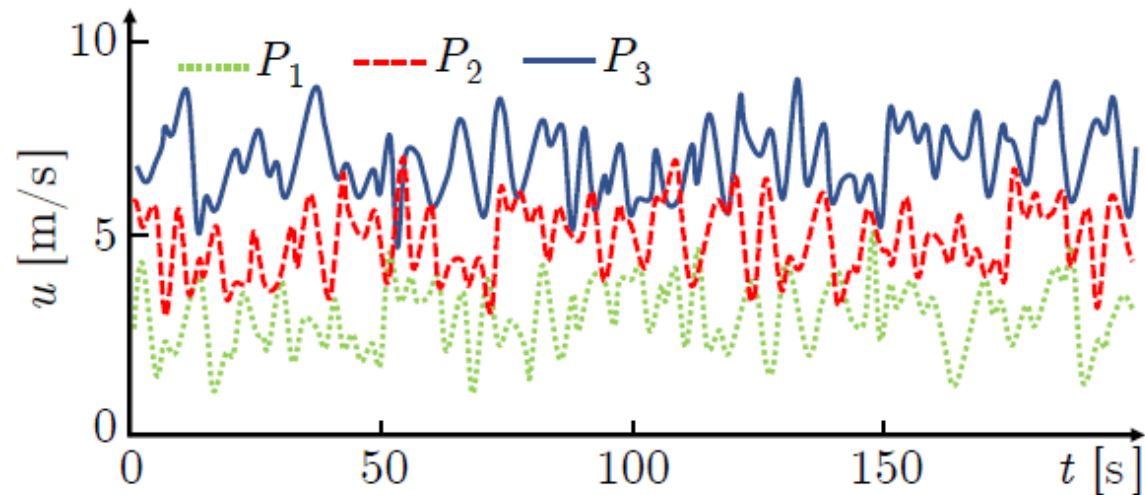
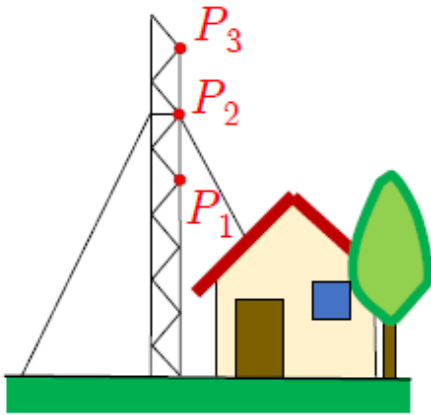


Tornadoes

It thus appears that we have numerous phenomena which characterise the wind climate of a specific location.

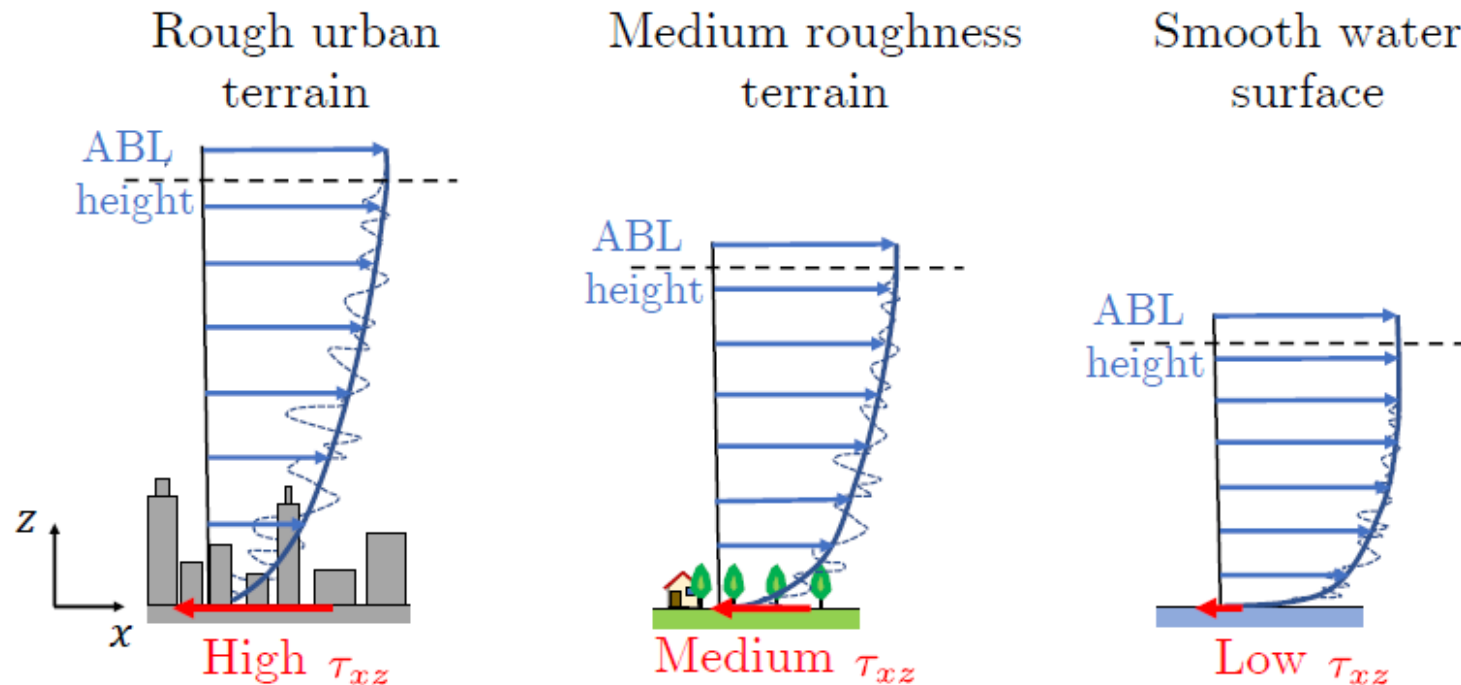
For design purposes, we mainly need to estimate a wind speed which has a sufficiently low probability to impinge on the structure.

Assume we measure the wind speed at different altitudes. The records will look as shown below. **The wind speed fluctuates continuously: it is a random variable.**



We can however recall the presence of the spectral gap and notice that a well-defined velocity, accounting only for the macro- and meso-scales, can be obtained taking a **10 min average** (or 1 hour average), $U_{10\text{min}}$ (or $U_{1\text{h}}$).

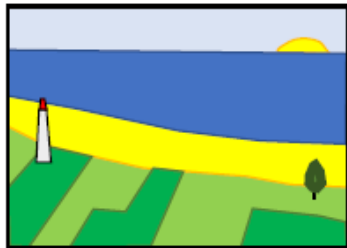
Depending on the roughness of the terrain, the variation of $U_{10\text{min}}$ with height changes, due to **the shear stress applied by the ground**.



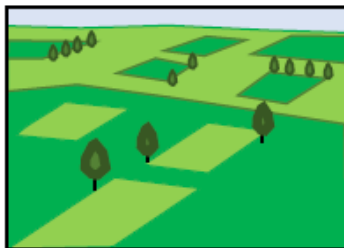
The Eurocode, for instance, provides the **variation of U_{10min} with the height** as

$$U_{10min}(z) = U_b k_r \ln(z/z_0),$$

where U_b is the base velocity, measured at 10 m height for Category II and evaluated basing on statistical analysis of historical measurements, while z_0 and k_r depend on the terrain Category.



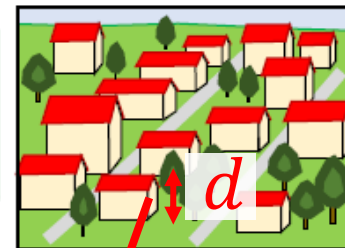
Category 0



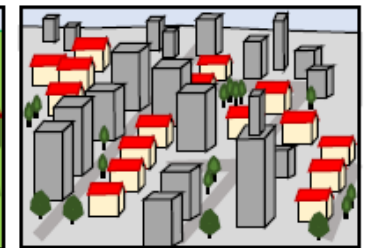
Category I



Category II



Category III



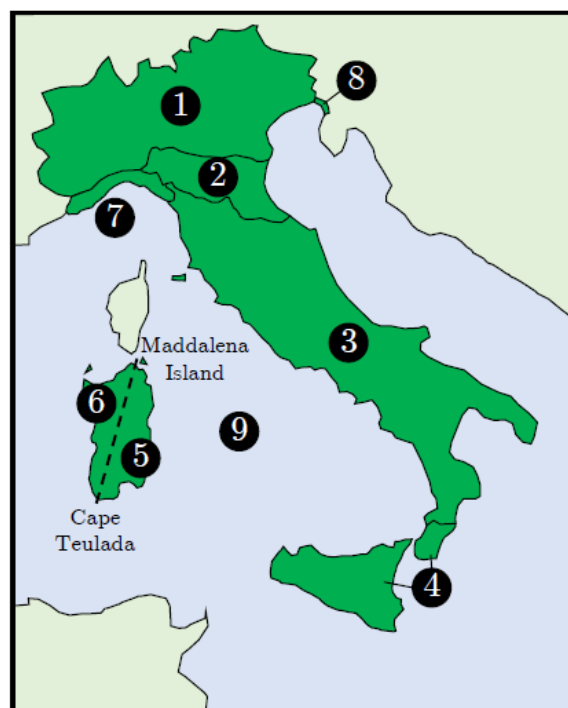
Category IV

Terrain category		z_0 [m]	z_{min} [m]	k_r [-]
0	Sea or coastal area exposed to the open sea	0.003	1.0	0.16
I	Lakes of flat and horizontal area with negligible vegetation and without obstacles	0.01	1.0	0.17
II	Area with low vegetation such as grass and isolated obstacles (trees, buildings) with separation of at least 20 obstacles heights	0.05	2.0	0.19
III	Area with regular cover of vegetation or buildings or with isolated obstacles with separation of maximum 20 obstacles heights (such as villages, suburban terrain, permanent forest)	0.3	5.0	0.22
IV	Area in which at least 15% of the surface is covered with buildings and their average height exceeds 15 m	1.0	10.0	0.24

$$d \approx [10 - 30] z_0.$$

$$k_r = 0.19 \left(\frac{z_0}{z_{0,II}} \right)^{0.07}$$

The base velocity at low altitude, based on historical data, is given by National Annexes and in the case of Italy



Zone	Description	$U_{b,0}$ [m/s]	a_0 [m]	k_a [-]
1	Valle d'Aosta, Piemonte, Lombardia, Trentino Alto Adige, Veneto, Friuli Venezia Giulia (with the exception of the province of Trieste)	25	1000	0.4
2	Emilia Romagna	25	750	0.45
3	Toscana, Marche, Umbria, Lazio, Abruzzo, Molise, Puglia, Campania, Basilicata, Calabria (excluding the province of Reggio Calabria)	27	500	0.37
4	Sicilia and the province of Reggio Calabria	28	500	0.36
5	Sardegna (zone on the east of the line joining Cape Teulada and the Maddalena island)	28	750	0.40
6	Sardegna (zone on the west of the line joining Cape Teulada and the Maddalena island)	28	500	0.36
7	Liguria	28	1000	0.54
8	Province of Trieste	30	1500	0.50
9	Islands (with the exception of Sicilia and Sardegna) and open sea	31	500	0.32

Corrections are provided to account for altitude.

We anticipated that wind speed is a **random variable**, both at the macro-meso scale and at the micro-scale.

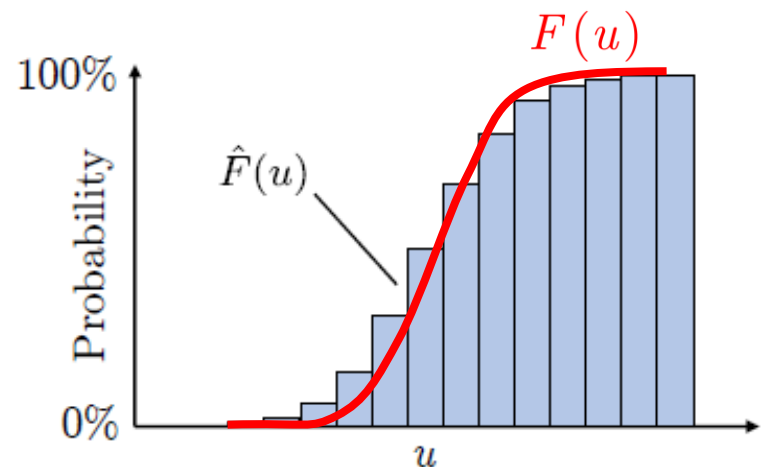
Let us assume to measure the instantaneous wind velocity at a point, say at 10 m from the ground, and repeat the measurement every 10 min. We collect results in a vector, denoted as **sample**

$$\hat{\mathbf{u}} = [\hat{u}(t_1), \hat{u}(t_2), \hat{u}(t_3), \dots, \hat{u}(t_n)],$$

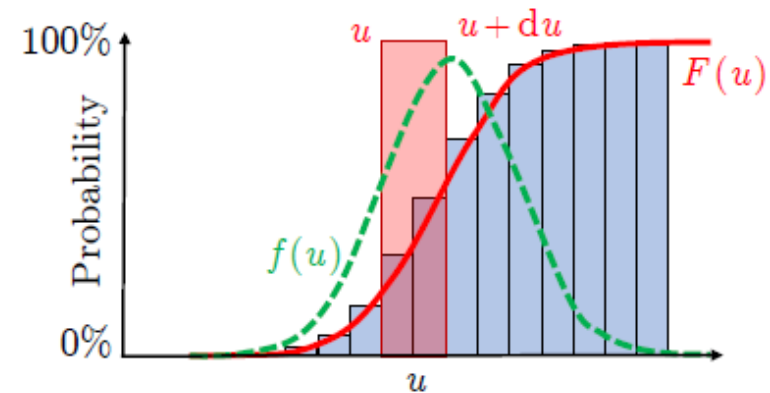
We denote the sample Cumulative Density Function, CDF, as

$$\hat{F}(u) = \frac{\#(\hat{\mathbf{u}} < u)}{\#(\hat{\mathbf{u}})},$$

In the limit $\# \hat{\mathbf{u}} \rightarrow \infty$ **the sample CDF tends to a well-defined function $F(u)$.**



It is usually of great interest to estimate the number of samples falling within a predefined range bounded by u and $u + du$.



We thus calculate

$$\#(\hat{\mathbf{u}} < (u + du)) - \#(\hat{\mathbf{u}} < u) = F(u + du)\#(\hat{\mathbf{u}}) - F(u)\#(\hat{\mathbf{u}}) = [F(u + du) - F(u)]\#(\hat{\mathbf{u}}),$$

Assuming du to be small, we can express $F(u)$ in Taylor series adopting a linear expansion

$$\#(\hat{\mathbf{u}} < (u + du)) - \#(\hat{\mathbf{u}} < u) = \left[F(u) + \frac{dF}{du}du - F(u) \right] \#(\hat{\mathbf{u}}) = f(u)du\#(\hat{\mathbf{u}}),$$

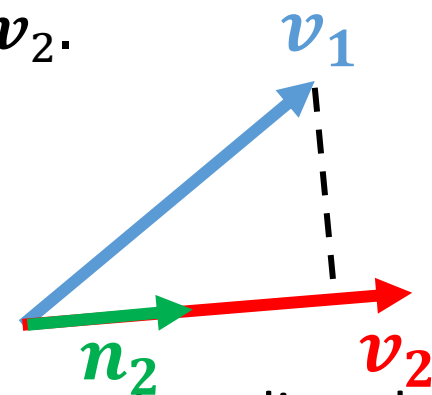
where $f(u) = \frac{dF}{du}$ is the **Probability Density Function**, PDF.

It is of interest for what follows to revise projections. In particular, we start by the usual vector projection, i.e. the calculation of a vector component. We consider two vectors in space $\mathbf{v}_1$ and $\mathbf{v}_2$.

Their scalar product reads

$$\mathbf{v}_1^T \mathbf{v}_2 = \mathbf{v}_1^T \mathbf{n}_2 |\mathbf{v}_2|$$

where $\mathbf{v}_1^T \mathbf{n}_2$ is the component of $\mathbf{v}_1$ in the direction pointed by $\mathbf{n}_2$. If $\mathbf{v}_2$ is a versor, $|\mathbf{v}_2| = 1$ and the scalar product directly provides the projection of $\mathbf{v}_1$ along $\mathbf{n}_2$.



A function can be seen as an infinite-dimensional vector, and the scalar product generalises to the L_2 projection

$$\mathcal{L}_2(v_1(u), v_2(u)) = \int_{-\infty}^{\infty} v_1(u)v_2(u)du.$$

If two vectors are characterized by the same projections in all directions they are identical.

As already stated, the PDF is a function, which can be thus considered as an infinitely dimensional vector.

A base of representation (we do not consider here its completeness) might be represented by monomials

$$u^j \text{ with } j = 1, 2, \dots, \infty,$$

The projections of $f(u)$ over the base of the monomials are

$$m_0 = \int_{-\infty}^{\infty} f(u)u^0 du = \int_{-\infty}^{\infty} f(u)du = 1.$$

$$m_1 = \int_{-\infty}^{\infty} f(u)u du = U, \quad (\text{Mean or Expected Value})$$

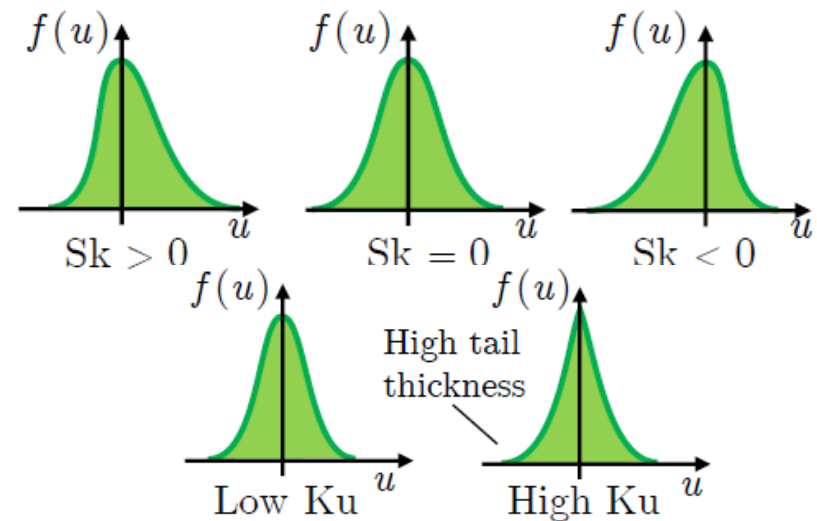
$$m_2 = \int_{-\infty}^{\infty} f(u)u^2 du = \text{RMS}^2(u), \quad (\text{Root Mean Square})$$

Usually, before proceeding to the calculation of the moments, the random variable is shifted by subtracting the mean, so obtaining the centered moments

$$m_{c2} = \int_{-\infty}^{\infty} f(u)(u - U)^2 du = \sigma_u^2,$$

$$m_{c3} = \int_{-\infty}^{\infty} f(u)(u - U)^3 du = \text{Sk}_u \sigma_u^3,$$

$$m_{c4} = \int_{-\infty}^{\infty} f(u)(u - U)^4 du = \text{Ku}_u \sigma_u^4,$$



σ_u^2 and σ_u are the variance and the standard deviation, while Sk_u and Ku_u are the skewness and kurtosis, respectively. **These quantities (for large J) can be estimated from the sample as**

$$\hat{U} = \frac{1}{J} \sum_{j=1}^J \hat{u}_j, \quad \hat{\sigma}_u^2 = \frac{1}{J} \sum_{j=1}^J (\hat{u}_j - \hat{U})^2,$$

$$\widehat{\text{Sk}}_u = \frac{1}{\hat{\sigma}_u^3 J} \sum_{j=1}^J (\hat{u}_j - \hat{U})^3, \quad \widehat{\text{Ku}}_u = \frac{1}{\hat{\sigma}_u^4 J} \sum_{j=1}^J (\hat{u}_j - \hat{U})^4.$$

Mean and fluctuating components CWE

For our purposes it will be often necessary to interpret the wind speed as the **superposition of its mean and its fluctuation around the mean**

$$u = U + u',$$

being as usual U the mean of u and u' the fluctuating part calculated as $u' = u - U$. We recall from the previous slide that

$$\sigma_u^2 = \int_{-\infty}^{\infty} f(u)(u - U)^2 du$$

so that

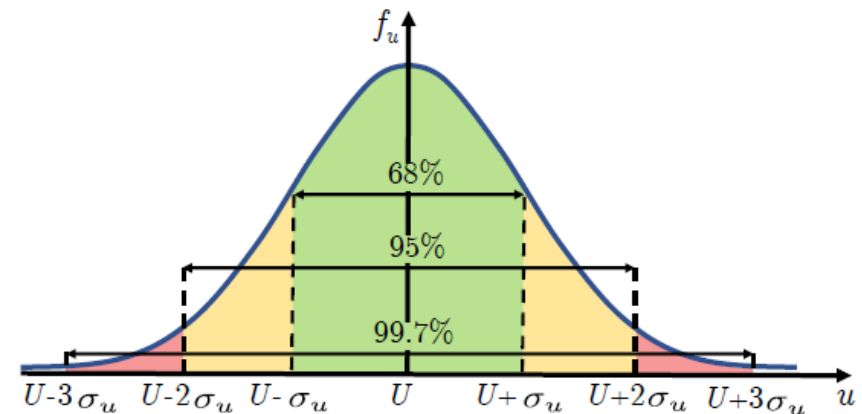
$$\sigma_u^2 = \sigma_{u'}^2$$

while

$$\text{RMS}^2(u) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N (\hat{u}_j)^2 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N (U + u'_j)^2 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N [U^2 + 2Uu'_j + (u'_j)^2] = U^2 + \sigma_u^2.$$

Often, knowing the generation mechanism of the random variable, it is possible to select *a priori* a parameterization of the statistical distribution.

For instance, by knowing that u is the sum of a multitude of independent random variables, none being preponderant, we known by the Central Limit Theorem that the distribution is Gaussian.



$$F(u) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{u - U}{\sigma_u \sqrt{2}} \right) \right] \quad f(u) = \frac{1}{\sqrt{2\pi\sigma_u^2}} e^{-\frac{1}{2} \left(\frac{u - U}{\sigma_u} \right)^2}.$$

Mean and standard deviation are enough to characterize the Gaussian distribution, while $\operatorname{Sk}_u = 0$ and $\operatorname{Ku}_u = 3$.

We now consider a random variable arising as the **weighted sum of two random variables**

$$v = \beta_1 u_1 + \beta_2 u_2$$

Its mean is

$$V = \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J \hat{v}_j = \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J (\beta_1 \hat{u}_{1,j} + \beta_2 \hat{u}_{2,j}) = \beta_1 U_1 + \beta_2 U_2,$$

while for the variance

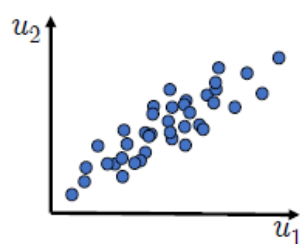
$$\begin{aligned} \sigma_v^2 &= \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J (\hat{v}_j - V)^2 \\ &= \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J [\beta_1 \hat{u}_{1,j} + \beta_2 \hat{u}_{2,j} - (\beta_1 U_1 + \beta_2 U_2)]^2 \\ &= \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J (\beta_1 \hat{u}'_{1,j} + \beta_2 \hat{u}'_{2,j})^2. \end{aligned}$$

expanding the square

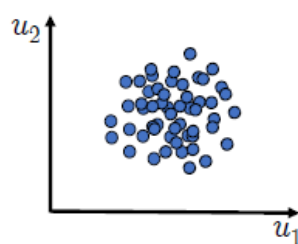
$$\begin{aligned} \sigma_v^2 &= \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J [(\beta_1 \hat{u}'_{1,j})^2 + 2\beta_1 \beta_2 \hat{u}'_{1,j} \hat{u}'_{2,j} + (\beta_2 \hat{u}'_{2,j})^2] \\ &= \beta_1^2 \sigma_1^2 + 2\beta_1 \beta_2 \operatorname{cov}(u_1, u_2) + \beta_2^2 \sigma_2^2, \end{aligned}$$

We thus define the **covariance between random variables** as

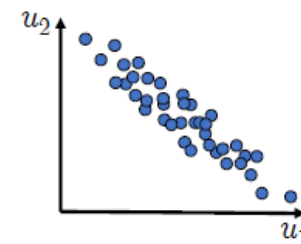
$$\text{cov}(u_1, u_2) = \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J \hat{u}'_{1,j} \hat{u}'_{2,j}.$$



$$\text{cov}(u_1, u_2) > 0$$



$$\text{cov}(u_1, u_2) \approx 0$$



$$\text{cov}(u_1, u_2) < 0$$

These expressions, defining $\beta = [\beta_1, \beta_2]$ and $U = [U_1, U_2]$ can be written

$$V = \beta^T U,$$

and, considering the variance

$$\sigma_v^2 = \beta^T \Sigma_u \beta, \quad \Sigma_u = \begin{bmatrix} \sigma_{u,1}^2 & \text{cov}(u_1, u_2) \\ \text{cov}(u_1, u_2) & \sigma_{u,2}^2 \end{bmatrix},$$

where Σ_u is the covariance matrix. **These expressions can be easily generalized to the sum of more than two random variables.**

Let us now consider **the sum of a random variable and noise**, i.e. a random variable we do not know much about.

$$v = \beta u + \epsilon.$$

We assume $\text{cov}(u, \epsilon) = 0$. Then, according to our previous formula

$$\sigma_v^2 = \beta^T \Sigma_u \beta = \beta^2 \sigma_u^2 + \sigma_\epsilon^2,$$

We can calculate

$$\begin{aligned} \text{cov}(v, u) &= \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J \hat{v}_j' \hat{u}_j' = \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J (\beta \hat{u}_j + \hat{\epsilon}_j)' \hat{u}_j' \\ &= \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J [(\beta \hat{u}_j + \hat{\epsilon}_j) - (\beta U + \mathcal{E})] \hat{u}_j' \\ &= \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J (\beta \hat{u}_j' + \hat{\epsilon}') \hat{u}_j' = \beta \sigma_u^2, \end{aligned}$$

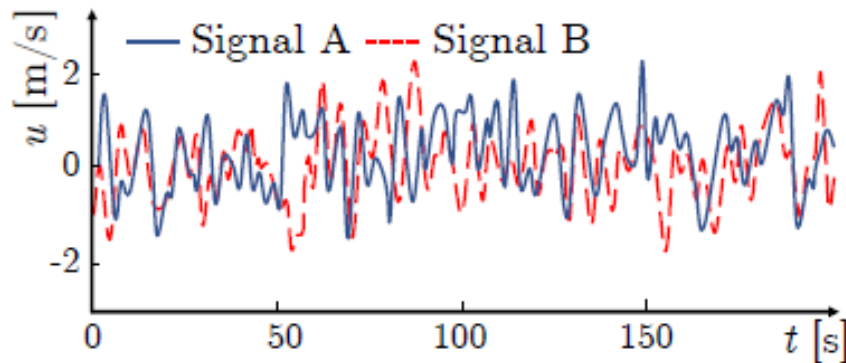
And finally define the correlation coefficient ρ_{vu}

$$\rho_{vu} = \frac{\text{cov}(v, u)}{\sigma_v \sigma_u} = \frac{\beta \sigma_u^2}{\sigma_u \sqrt{\beta^2 \sigma_u^2 + \sigma_\epsilon^2}} = \frac{\beta \sigma_u^2}{\sigma_u^2 \sqrt{\beta^2 + \sigma_\epsilon^2 / \sigma_u^2}} = \frac{\beta}{|\beta|} \frac{1}{\sqrt{1 + \sigma_\epsilon^2 / (\beta^2 \sigma_u^2)}}.$$

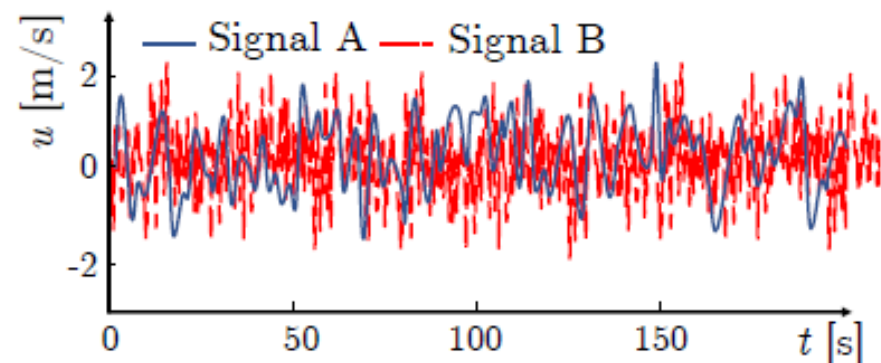
We now go back to our original problem, i.e. characterizing velocity time histories from the statistical point of view. In principle we shall provide the PDF. However, wind velocities are usually Gaussian so that we can provide U and σ_u , only. Usually, we adopt

$$I_u = \frac{\sigma_u}{U},$$

where I_u is the turbulence intensity. If $I_u = 0.1$ it means that the typical velocity fluctuation is 10% of the mean value. Keeping I_u constant, we might have



CASE A

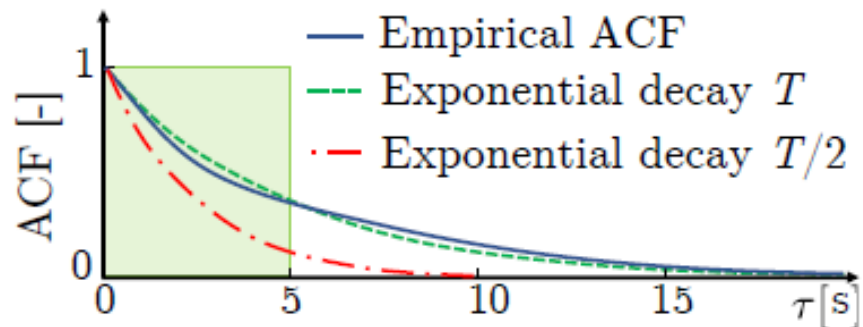


CASE B

The two cases are obviously different, In CASE 1 the two signal are different but they are more comparable than CASE 2 (the red signal seems “quicker” in fluctuating). We can assume the signal to be built as

$$u(t + \tau) = \beta u(t) + \epsilon,$$

We can calculate the correlation coefficient between the signals at two different times $\rho_{u(t), u(t+\tau)}$. The result usually looks as $\rho_{uu}(\tau) = e^{-\tau/T}$

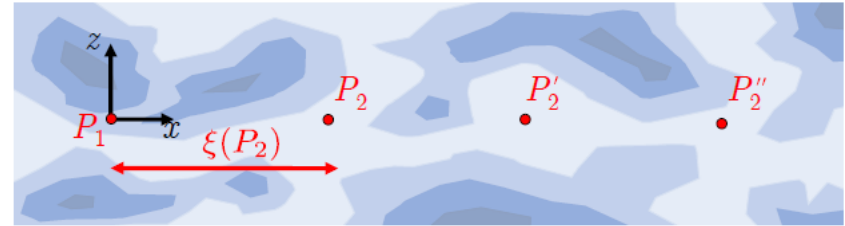


We thus define the **integral time scale** as $T_u = \int_0^{\infty} \rho_{uu}(\tau) d\tau$. It can be easily shown that T_u is exactly the coefficient appearing in the exponential approximation above.

The same considerations can be made in space

$$L_u^x = \int_0^\infty \rho(u(0), u(\xi)) d\xi,$$

so defining the integral length scales.



Considering all velocity components and all spatial directions, we arrive to

$$\begin{bmatrix} T_u \\ T_v \\ T_w \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} L_u^x & L_u^y & L_u^z \\ L_v^x & L_v^y & L_v^z \\ L_w^x & L_w^y & L_w^z \end{bmatrix}$$

and it can be shown that the following are approximately true

$$L_u^x = T_u U, \quad L_v^x = T_v U, \quad L_w^x = T_w U.$$

In general, when we represent the velocity field in the Atmospheric Boundary Layer

$$\mathbf{u}(\mathbf{x}, t) = [u(\mathbf{x}, t), v(\mathbf{x}, t), w(\mathbf{x}, t)]^T$$

The mean field has been already discussed

$$\mathbf{U}_{10min}(z) = [U_{10min}(z), 0, 0]$$

As regard the fluctuating part

$$\sigma_u = k_r U_b, \quad \sigma_v = 0.75\sigma_u, \quad \sigma_w = 0.5\sigma_u.$$

According to previous formula, **the turbulence intensity profile is**

$$I_u = \frac{\sigma_u}{U(z)} = \frac{k_r U_b}{U_b k_r \ln(z/z_0)} = \frac{1}{\ln(z/z_0)},$$

while, very roughly

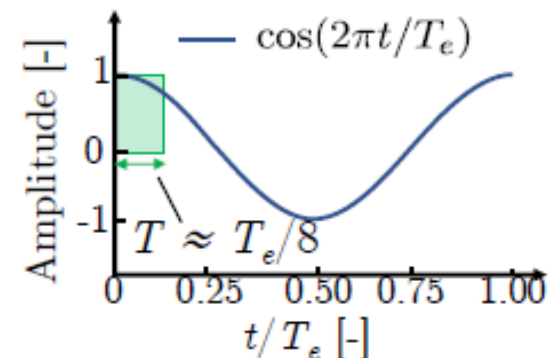
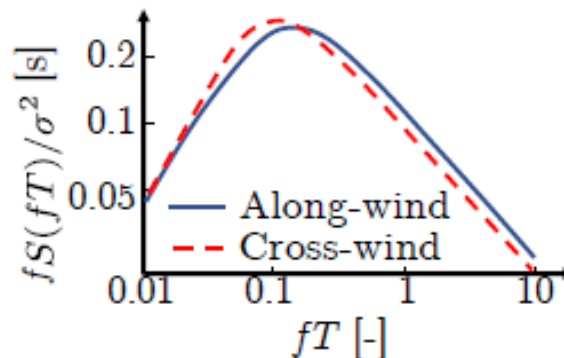
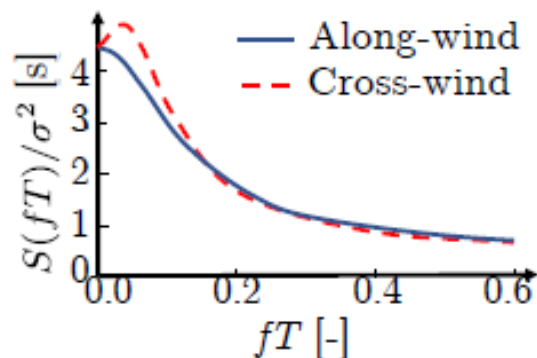
$$L_u^x = 300 \left(\frac{z}{200} \right)^{0.67+0.05\ln(z_0)} \cdot \begin{bmatrix} L_u^x & L_u^y & L_u^z \\ L_v^x & L_v^y & L_v^z \\ L_w^x & L_w^y & L_w^z \end{bmatrix} = L_u^x \begin{bmatrix} 1.0 & 0.6 & 0.4 \\ 0.4 & 0.4 & 0.3 \\ 0.3 & 0.3 & 0.2 \end{bmatrix}$$

Within the ABL, turbulence is usually characterized by spectra which can be approximated by empirical formulations such as **the von Karman spectra**

$$S_u(f) = \sigma_u^2 \frac{4T_u}{[1 + 70.8(fT_u)^2]^{5/6}},$$

$$S_v(f) = \sigma_v^2 \frac{4T_v [1 + 188.4(2fT_v)^2]}{[1 + 70.8(2fT_v)^2]^{11/6}},$$

$$S_w(f) = \sigma_w^2 \frac{4T_w [1 + 188.4(2fT_w)^2]}{[1 + 70.8(2fT_w)^2]^{11/6}},$$



where we recognize again the integral time-scales.